LINUX ASSIGNMENT

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Command line arguments, fileoperators, read and filters

Write a Bash script that accepts a filename as the first command-line argument.

Check whether an argument is provided.

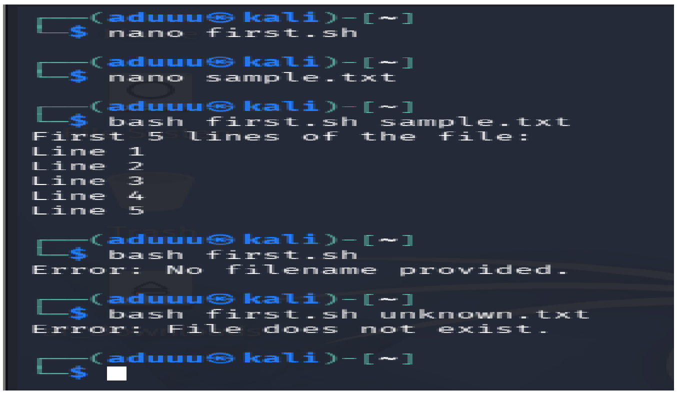
Check whether the file exists using a file operator.

If the file exists, display the first 5 lines using head.

Otherwise, display an appropriate error message.

File Type and Permission Checker

```
#!/bin/bash
if [ -z "$1" ]
then
    echo "Error: No filename provided."
    exit 1
fi
if [ -e "$1" ]
then
    echo "First 5 lines of the file:"
    head -n 5 "$1"
else
    echo "Error: File does not exist."
fi
```



Write a Bash script that accepts a path as a command-line argument and determines whether it is:

a regular file,

a directory,

readable,

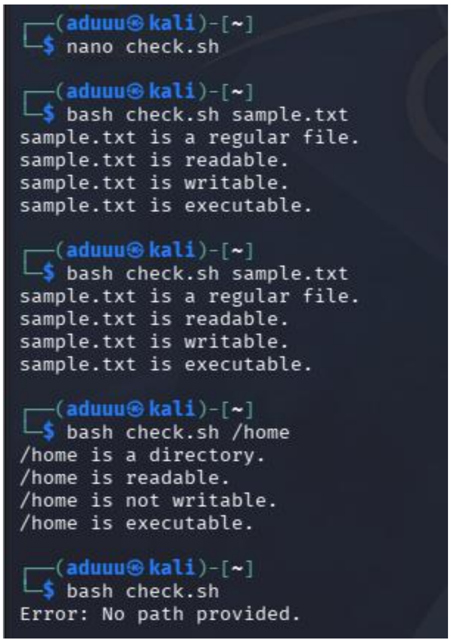
writable, and

executable.

Display suitable messages using file operators such as -f, -d, -r, -w, and -x.

```
#!/bin/bash
if [ -z "$1" ]
then
   echo "Error: No path provided."
   exit 1
fi
if [ -f "$1" ]
then
   echo "$1 is a regular file."
elif [ -d "$1" ]
then
   echo "$1 is a directory."
else
   echo "$1 is neither a regular file nor a directory."
fi
if [ -r "$1" ]
then
   echo "$1 is readable."
else
   echo "$1 is not readable."
fi
if [ -w "$1" ]
then
   echo "$1 is writable."
```

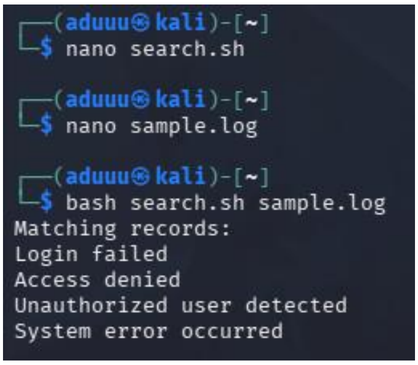
```
else
    echo "$1 is not writable."
fi
if [ -x "$1" ]
then
    echo "$1 is executable."
else
    echo "$1 is not executable."
fi
```



Write a Bash script that accepts a log file as a command-line argument and searches for the words: failed, denied, unauthorized, and error.

```
#!/bin/bash
if [ -z "$1" ]
then
    echo "Error: No log file provided."
    exit 1
fi
if [ ! -f "$1" ]
then
```

```
    echo "Error: Log file does not exist."
    exit 1
fi
echo "Matching records:"
grep -iE "failed|denied|unauthorized|error" "$1"
```



Write a Bash script that accepts a filename and checks whether:

the file exists,

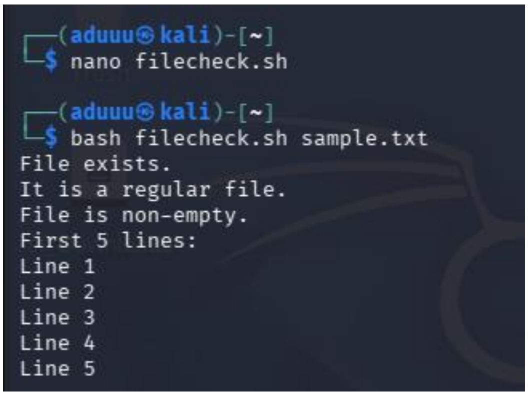
it is a regular file,

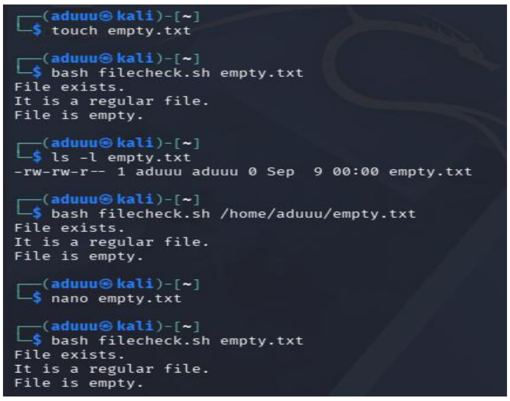
it is empty or non-empty.

If it is non-empty, display the first 5 lines.

```
#!/bin/bash
if [ -z "$1" ]
then
    echo "Error: No filename provided."
    exit 1
fi
if [ ! -e "$1" ]
then
    echo "Error: File does not exist."
    exit 1
fi
```

```
echo "File exists."
if [ -f "$1" ]
then
   echo "It is a regular file."
else
   echo "It is not a regular file."
   exit 1
fi
if [ -s "$1" ]
then
   echo "File is non-empty."
   echo "First 5 lines:"
   head -n 5 "$1"
else
   echo "File is empty."
fi
```





Explanation of the operators:

-e Checks whether the file exists

-f Checks whether the path is a regular file

-s Checks whether the file is non-empty

head -n 5 Displays the first five lines

Write a Bash script named investigate.sh that accepts a log file as a command-line argument.

The script must first verify that the file exists and is readable. Then ask the investigator to enter a search term.

Display the first 5 matching records and report the total number of matching records.

```
#!/bin/bash
if [ -z "$1" ]
then
   echo "Error: No log file provided."
   exit 1
fi
if [ ! -e "$1" ]
then
   echo "Error: Log file does not exist."
   exit 1
fi
if [ ! -r "$1" ]
then
   echo "Error: Log file is not readable."
   exit 1
fi
read -p "Enter search term: " term
echo "First 5 matching records:"
grep -i "$term" "$1" | head -n 5
count=$(grep -ic "$term" "$1")
echo "Total matching records: $count"
```